Erik Hogset

Mechanical Engineering Senior | Design, Energy Systems, and Manufacturing
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Portfolio: erikhogset.github.io

Education

Boston University College of Engineering, Boston, MA — Expected May 2026
B.S. in Mechanical Engineering
Relevant Coursework: Manufacturing Processes • Automation & Manufacturing Methods • Additive Manufacturing • Electromechanical Design • Mechanics of Materials • Senior Design

Technical Skills

Manufacturing: CNC Milling, CNC Turning, Manual Milling, Manual Turning, 3D Printing (FFF, SLA), TIG Welding, CO_2 Laser Cutter, Waterjet Cutting, Sheet Metal Fabrication
CAD / CAM: SolidWorks, Autodesk Fusion 360, Onshape
Programming / Controls: MATLAB, LabVIEW, Python, C, Data Acquisition (DAQ), Signal Processing

Work Experience

Engineering Product Innovation Center (E.P.I.C.), Boston, MA. — May 2024 - Present
Laboratory Assistant

Supported prototyping and fabrication for engineering projects in a high-traffic makerspace
Operated CNC mills, CNC lathes, manual mills/lathes, and 3D printers (*FFF* and *SLA*)
Generated CAM toolpaths and G-code using SolidWorks and Autodesk Fusion 360
Assisted with part fixturing, tolerancing, and design adjustments for manufacturability
Enforced lab safety and proper machine operation while supporting multiple users

Starbucks, Breezewood, PA and Boston, MA — Jun. 2021 – Sep. 2025
Barista

Balanced full-time Mechanical Engineering coursework with part-time employment
Worked in a fast-paced, team-based environment requiring accuracy, multitasking, and efficiency

Projects

Reactive Road Sign System – Senior Design I & II (*ME460/461*)
- Designed a retrofit mechanical system for roadside signposts to deflect under extreme wind loads
- Evaluated joint concepts using CAD, materials selection, and structural failure analysis
- Modeled wind-induced loading and bending moments to size a sacrificial breakaway pin
- Built and tested physical prototypes to validate loading behavior and manufacturability

Compact Fishing Kit Product – Automation & Manufacturing Methods (*ME345*)
- Designed a consumer product and developed a CAD-to-CAM workflow for CNC manufacturing
- Created CNC toolpaths and programmed collaborative robots for machine tending and assembly
- Integrated conveyors, robot programs, and CNC machines into an automated manufacturing cell

Electromechanical Interactive Ouija Board – Electromechanical Design (*ME360*)
- Built a 2-axis, belt-driven motion system to position a planchette using stepper motors
- Implemented motion control to convert AI-generated responses into coordinate targets
- Developed control software integrating voice input, AI API communication, and motor control

Adaptive LED Lamp – Introduction to Engineering Design (*EK210*)
- Designed and built a motion-activated LED lamp using electronic sensing and physical prototyping
- Implemented ambient-light sensing and time-based control to reduce blue-light exposure